

GREEN'S OPEN PROBLEM 51 AT DENSITY ONE HALF: BOUNDED CODIMENSION IN BINARY SUMSETS

JENSEN KOHLMAYER AND LIAM KRUER

ABSTRACT. We prove that if $A \subseteq \mathbb{F}_2^n$ has density at least $1/2 - \varepsilon$, where $0 \leq \varepsilon \leq 1/4$, then $A + A$ contains a linear subspace of codimension at most $1 + 256e n \varepsilon^2$. Consequently, density at least $1/2 - k/\sqrt{n}$ guarantees a subspace of codimension at most $\lceil 1 + 256ek^2 \rceil$ for all sufficiently large n . This answers the one-half-density question recorded under Green's Problem 51, and Question 5.1 of Sanders. The proof combines an elementary variance estimate for ternary separators of the Boolean cube, averaging over random coordinate faces, and polynomial interpolation on a minimal covering obstruction. A complete Lean 4 formalization of the specified Formal Conjectures target accompanies the paper.

1. INTRODUCTION

Let $G = \mathbb{F}_2^n$, and write $\text{dens}(A) = |A|/2^n$ for the density of a set $A \subseteq G$. We consider the dimension of a subspace contained in the sumset

$$A + A = \{a + a' : a, a' \in A\}.$$

Green's Problem 51 asks, more generally, for the largest size of an affine subspace guaranteed in a sumset of a given density [2]. Its one-half-density question asks whether, for each fixed $k > 0$, density at least $1/2 - k/\sqrt{n}$ guarantees codimension bounded independently of n . Sanders posed the corresponding question for linear subspaces in [3, Question 5.1]. We prove the following quantitative result.

Theorem 1.1. *Let $n \geq 0$ be an integer and $0 \leq \varepsilon \leq 1/4$. If $A \subseteq \mathbb{F}_2^n$ satisfies $\text{dens}(A) \geq 1/2 - \varepsilon$, then there is a linear subspace $H \leq \mathbb{F}_2^n$ such that*

$$H \subseteq A + A, \quad \text{codim } H \leq 1 + 256e n \varepsilon^2. \quad (1)$$

Here $e = \exp(1)$, and codimension means $n - \dim H$.

Corollary 1.2. *Fix $k > 0$, and set*

$$c_k = \lceil 1 + 256ek^2 \rceil, \quad N_k = \lceil 16k^2 \rceil + 1. \quad (2)$$

For every integer $n \geq N_k$ and every $A \subseteq \mathbb{F}_2^n$ of density at least $1/2 - k/\sqrt{n}$, the sumset $A + A$ contains a linear subspace of codimension at most c_k .

When $\text{dens}(A) > 1/2$, the sets A and $x + A$ intersect for every $x \in G$, so $A + A = G$. Sanders proved that codimension one is still guaranteed when the density falls below $1/2$ by a sufficiently small absolute multiple of $n^{-1/2}$ [3, Theorem 1.2]. Corollary 1.2 allows every fixed multiple, with a quadratic bound on the codimension. Our result addresses this specified part of Green's Problem 51; it does not determine the optimal guaranteed dimension at every density. No optimality is asserted for the constant $256e$ or for the dependence on k .

Date: September 2026.

2020 Mathematics Subject Classification. Primary 11B13; Secondary 05D05.

Key words and phrases. Sumsets, vector spaces over finite fields, Boolean cube, Walsh transform, polynomial interpolation, Lean formalization.

The argument has two parts. First, for every nonempty subset D of the missing sums $G \setminus (A + A)$, we find a linear functional that vanishes on at most a $256\epsilon^2$ fraction of D . To do this, we lift a difference of indicators to a ternary function on a cube whose coordinates are indexed by D . A separator estimate and an average over random faces force a nonzero Walsh coefficient of small degree. That coefficient identifies the required functional. Second, polynomial interpolation bounds the size of a minimal obstruction to covering the missing sums by complements of kernels. Combining this bound with repeated application of the first part removes the logarithmic loss that a direct greedy argument on the whole set would incur.

All vector spaces and linear functionals in the combinatorial argument are over $\mathbb{F}_2$. Expectations on finite sets use the uniform probability measure unless another distribution is specified. The analytic statements use real-valued functions. Sections 2–6 give a self-contained proof; Section 7 describes the accompanying formalization.

2. A VARIANCE BOUND FOR CUBE SEPARATORS

Write $Q_m = \{0, 1\}^m$, with two vertices adjacent when they differ in exactly one coordinate. A function $u : Q_m \rightarrow \{-1, 0, 1\}$ is a *ternary separator* if no edge joins a vertex with value 1 to a vertex with value -1 . Put

$$d(u) = \mathbb{P}_x[u(x) = 0], \quad \text{Var}(u) = \mathbb{E}_x u(x)^2 - (\mathbb{E}_x u(x))^2.$$

Thus $\mathbb{E}u^2 = 1 - d(u)$. The next estimate needs no isoperimetric theorem.

Lemma 2.1. *Every ternary separator on Q_m satisfies*

$$\text{Var}(u) \leq 2\sqrt{m} d(u). \quad (3)$$

Proof. We induct on m . For $m = 0$ the cube consists of one point and the variance is zero. Suppose $m \geq 1$, and let u_0, u_1 be the restrictions obtained by fixing the last coordinate. Write $d_i = d(u_i)$ and $\mu_i = \mathbb{E}u_i$, so that $d = (d_0 + d_1)/2 = d(u)$. The separator condition gives, for each $x \in Q_{m-1}$,

$$|u_0(x) - u_1(x)| \leq \mathbf{1}_{\{u_0(x)=0\}} + \mathbf{1}_{\{u_1(x)=0\}}.$$

Indeed, if neither value is zero they must agree. Averaging yields $|\mu_0 - \mu_1| \leq 2d$. The variance decomposition and the induction hypothesis therefore give

$$\text{Var}(u) = \frac{\text{Var}(u_0) + \text{Var}(u_1)}{2} + \frac{(\mu_0 - \mu_1)^2}{4} \leq 2\sqrt{m-1} d + d^2. \quad (4)$$

If $\sqrt{m} d \geq 1$, then $\text{Var}(u) \leq \mathbb{E}u^2 \leq 1 \leq 2\sqrt{m} d$. Otherwise,

$$d < \frac{1}{\sqrt{m}} \leq \frac{2}{\sqrt{m} + \sqrt{m-1}} = 2(\sqrt{m} - \sqrt{m-1}).$$

Substitution in (4) proves (3). $\square$

The same estimate applies to every coordinate face of the cube. Averaging those face estimates at a suitable scale will give information about the degree of a nonzero Fourier coefficient.

3. RANDOM FACES AND A LOW-DEGREE WALSH COEFFICIENT

For $S \subseteq [m] = \{1, \dots, m\}$, let

$$\chi_S(x) = (-1)^{\sum_{i \in S} x_i}, \quad \widehat{u}(S) = \mathbb{E}_x u(x) \chi_S(x).$$

The characters form an orthonormal basis for real functions on Q_m . Indeed, $\mathbb{E}\chi_S\chi_T$ is 1 if $S = T$ and is zero otherwise, by averaging in any coordinate of $S \triangle T$. Since there are 2^m characters, orthogonality gives both inversion and Parseval:

$$u(x) = \sum_{S \subseteq [m]} \widehat{u}(S) \chi_S(x), \quad \mathbb{E}u^2 = \sum_{S \subseteq [m]} \widehat{u}(S)^2. \quad (5)$$

Fix $J \subseteq [m]$. Let $V_J(u)$ be the average, over all assignments to the coordinates outside J , of the variance of the restriction of u to the resulting J -face. Averaging a character over the coordinates in J kills it exactly when $S \cap J \neq \emptyset$. Consequently, the identity $V_J(u) = \mathbb{E}u^2 - \mathbb{E}(\mathbb{E}[u \mid x_{[m] \setminus J}])^2$ and Parseval give

$$V_J(u) = \sum_{S \subseteq [m]} \widehat{u}(S)^2 \mathbf{1}_{\{S \cap J \neq \emptyset\}}. \quad (6)$$

If u is a ternary separator, Lemma 2.1 applied on each face gives

$$V_J(u) \leq 2\sqrt{|J|} d(u), \quad (7)$$

because the average zero density of the faces is $d(u)$.

Now choose J by including each coordinate independently with probability $\theta \in [0, 1]$. Taking expectations in (6) and (7), and using Cauchy–Schwarz, yields

$$\sum_{S \subseteq [m]} \widehat{u}(S)^2 (1 - (1 - \theta)^{|S|}) \leq 2d(u) \mathbb{E}_J \sqrt{|J|} \leq 2d(u) \sqrt{m\theta}. \quad (8)$$

Lemma 3.1. *If $u : Q_m \rightarrow \{-1, 0, 1\}$ is a ternary separator and $d = d(u) \leq 1/2$, then there is $S \subseteq [m]$ with*

$$\widehat{u}(S) \neq 0, \quad |S| \leq 64md^2. \quad (9)$$

Proof. Since $\mathbb{E}u^2 = 1 - d > 0$, Parseval guarantees a nonzero coefficient. Let s be the smallest degree of such a coefficient. If $s = 0$, the conclusion holds. Suppose $s \geq 1$, and take $\theta = 1/s$ in (8).

For $0 \leq \theta \leq 1$ and every integer $j \geq 0$, the elementary inequality

$$(1 - \theta)^j (1 + j\theta) \leq 1 \quad (10)$$

follows by induction: in the induction step use $(1 - \theta)(1 + (j + 1)\theta) = 1 + j\theta - (j + 1)\theta^2 \leq 1 + j\theta$. In particular, $(1 - 1/s)^s \leq 1/2$. For every coefficient that is nonzero we have $|S| \geq s$, and hence $1 - (1 - 1/s)^{|S|} \geq 1/2$. Thus (8) implies

$$\frac{1 - d}{2} \leq 2d\sqrt{m/s}.$$

Both sides are nonnegative. Squaring and multiplying by $4s$ gives $s(1 - d)^2 \leq 16md^2$. Since $1 - d \geq 1/2$, we obtain $s \leq 64md^2$, as required. $\square$

4. A LINEAR FUNCTIONAL ON MISSING SUMS

The dual space $G^* = \text{Hom}_{\mathbb{F}_2}(G, \mathbb{F}_2)$ indexes the characters $\chi_\gamma(x) = (-1)^{\gamma(x)}$ of G . Exponents in this notation take the representatives 0, 1 of elements of $\mathbb{F}_2$. Fourier inversion on G is the same orthogonality argument as (5).

Proposition 4.1. *Suppose $0 \leq \varepsilon \leq 1/4$ and $\text{dens}(A) \geq 1/2 - \varepsilon$. For every nonempty $D \subseteq G \setminus (A + A)$ there exists $\gamma \in G^*$ such that*

$$|D \cap \ker \gamma| \leq 256\varepsilon^2 |D|. \quad (11)$$

Proof. Choose $b_0 \in D$, write $\alpha = \text{dens}(A)$, and define

$$h(x) = \mathbf{1}_A(x) - \mathbf{1}_A(x + b_0).$$

As $b_0 \notin A + A$, the sets A and $A + b_0$ are disjoint. Therefore h takes values in $\{-1, 0, 1\}$ and

$$\mathbb{P}_x[h(x) = 0] = 1 - 2\alpha \leq 2\varepsilon. \quad (12)$$

For every $b \in D$ and $x \in G$ we also have

$$h(x)h(x + b) \neq 1. \quad (13)$$

If both values were 1, then $x, x + b \in A$, which would give $b \in A + A$. If both were -1 , the same contradiction would follow from $x + b_0, x + b + b_0 \in A$.

Enumerate $D = \{b_1, \dots, b_m\}$, and let

$$\pi : Q_m \longrightarrow G, \quad \pi(z) = \sum_{i=1}^m z_i b_i.$$

For $t \in G$, define the lift

$$u_t(z) = (-1)^{z_1 + \dots + z_m} h(t + \pi(z)). \quad (14)$$

This is a ternary separator. In fact, flipping coordinate i changes the parity factor's sign and translates the argument of h by b_i ; opposite nonzero values of u_t across that edge would force the product in (13) to equal 1.

For each fixed z , the point $t + \pi(z)$ is uniform in G when t is uniform. Thus, by averaging (12),

$$\mathbb{E}_t d(u_t) = \mathbb{P}_x[h(x) = 0] \leq 2\varepsilon.$$

Choose t with $d_t = d(u_t) \leq 2\varepsilon \leq 1/2$. Lemma 3.1 supplies a nonzero Walsh coefficient of u_t at some set S satisfying

$$|S| \leq 64md_t^2 \leq 256m\varepsilon^2. \quad (15)$$

For $\gamma \in G^*$, set

$$S_\gamma = \{i \in [m] : \gamma(b_i) = 0\}.$$

Fourier inversion for h , substituted in (14), gives the exact identity

$$u_t(z) = \sum_{\gamma \in G^*} \widehat{h}(\gamma) \chi_\gamma(t) \chi_{S_\gamma}(z). \quad (16)$$

Indeed, the exponent contributed by coordinate z_i is $(1 + \gamma(b_i))z_i$, which is z_i precisely when $\gamma(b_i) = 0$. Orthogonality now shows that a nonzero coefficient at S can occur only if $S = S_\gamma$ for at least one γ . This conclusion does not require D to span G : coinciding frequencies in (16) can cancel, but cannot produce a frequency absent from the displayed sum. For this γ , we have $|D \cap \ker \gamma| = |S|$, and (15) proves (11). $\square$

5. CRITICAL OBSTRUCTIONS AND POLYNOMIAL INTERPOLATION

For $B \subseteq G \setminus \{0\}$, define $c(B)$ to be the least nonnegative integer r for which there are $\gamma_1, \dots, \gamma_r \in G^*$ with

$$B \cap \bigcap_{j=1}^r \ker \gamma_j = \emptyset. \quad (17)$$

The empty intersection is G , so $c(\emptyset) = 0$. Coordinate functionals give $c(B) \leq n$. Equivalently, $c(B)$ is the least codimension of a linear subspace avoiding B : the common kernel in (17) has codimension at most r , and every subspace of codimension r is the common kernel of r independent linear functionals.

Lemma 5.1. *Let $r \geq 0$. If $D \subseteq G$ is inclusion-minimal among sets that cannot be covered by r complements of kernels, then*

$$|D| \leq M(n, r), \quad M(n, r) = \sum_{j=0}^{\min(r, n)} \binom{n}{j}. \quad (18)$$

Proof. For every $b \in D$, the proper subset $D \setminus \{b\}$ is coverable by r complements of kernels. Choose corresponding functionals $\gamma_{b,1}, \dots, \gamma_{b,r}$. Each of these functionals must vanish at b ; otherwise the same collection would cover all of D . Define the $\mathbb{F}_2$ -valued polynomial function

$$P_b(x) = \prod_{j=1}^r (1 + \gamma_{b,j}(x)). \quad (19)$$

For $x \in D$, the value is 1 if $x = b$ and is 0 otherwise. The functions P_b , $b \in D$, are therefore linearly independent: evaluation at each point of D isolates its coefficient in any linear relation.

As a formal polynomial, P_b has degree at most r . As a function on $\mathbb{F}_2^n$, it can be reduced using $x_i^2 = x_i$ to a linear combination of the squarefree monomials $x_S = \prod_{i \in S} x_i$ with $|S| \leq r$. There are $M(n, r)$ such monomials, so the space of their linear combinations has dimension at most $M(n, r)$. It contains the independent family $\{P_b : b \in D\}$, proving (18). The same argument includes $r = 0$, with empty products equal to 1. $\square$

Lemma 5.2. *For integers $1 \leq r \leq n$,*

$$\left(\frac{r}{n}\right)^r M(n, r) \leq e^r. \quad (20)$$

Proof. Put $t = r/n \in (0, 1]$. For $|S| \leq r$, one has $t^r \leq t^{|S|}$. Summing over subsets and then using $1 + t \leq e^t$ gives

$$t^r M(n, r) \leq \sum_{S \subseteq [n]} t^{|S|} = (1 + t)^n \leq e^{nt} = e^r. \quad \square$$

Proposition 5.3. *Let $B \subseteq G \setminus \{0\}$ and $\delta \geq 0$. Suppose that for every nonempty $D \subseteq B$ there is $\gamma \in G^*$ with*

$$|D \cap \ker \gamma| \leq \delta |D|. \quad (21)$$

Then

$$c(B) \leq 1 + en\delta. \quad (22)$$

Proof. If $c(B) \leq 1$, the conclusion is immediate. Otherwise write $c(B) = r + 1$, where $1 \leq r \leq n$. Choose an inclusion-minimal $D \subseteq B$ that cannot be covered by r complements of kernels. Such a set exists because B is finite and is itself not coverable by r of them.

Starting with $D_0 = D$, apply (21) successively to obtain

$$D_j = D_{j-1} \cap \ker \gamma_j, \quad |D_j| \leq \delta |D_{j-1}| \quad (1 \leq j \leq r).$$

All these sets are nonempty, since otherwise at most r complements of kernels would cover D ; extra zero functionals can pad a shorter collection to length r . Consequently

$$1 \leq |D_r| \leq \delta^r |D| \leq \delta^r M(n, r), \quad (23)$$

where Lemma 5.1 gives the last inequality. Multiplying by $(r/n)^r$ and applying Lemma 5.2 yields

$$(r/n)^r \leq \delta^r (r/n)^r M(n, r) \leq (e\delta)^r.$$

Both bases are nonnegative and $r \geq 1$, so taking r th roots gives $r \leq en\delta$. Therefore $c(B) = r + 1 \leq 1 + en\delta$. $\square$

The minimality of D is essential to this estimate: it supplies a polynomial that separates each point of D from all the other points. This bounds the size of the obstruction in terms of its covering parameter, rather than in terms of $|B|$, and is what makes (22) effective.

6. COMPLETION OF THE SUMSET ARGUMENT

Proof of Theorem 1.1. Set $B = G \setminus (A + A)$. The density hypothesis gives $\text{dens}(A) \geq 1/4 > 0$, so A is nonempty. Since $a + a = 0$ for every $a \in A$, we have $0 \notin B$. Proposition 4.1 verifies the hypothesis of Proposition 5.3 with $\delta = 256\varepsilon^2$. It follows that

$$c(B) \leq 1 + 256e n \varepsilon^2.$$

Choose functionals witnessing $c(B)$ and let H be their common kernel. Then H avoids B , so $H \subseteq A + A$. Rank-nullity applied to the map $x \mapsto (\gamma_1(x), \dots, \gamma_{c(B)}(x))$ gives $\text{codim } H \leq c(B)$. This proves (1), including the case $B = \emptyset$, when $H = G$. $\square$

Proof of Corollary 1.2. If $n \geq N_k$, then $n > 16k^2 > 0$. Put $\varepsilon = k/\sqrt{n}$. We have $0 < \varepsilon < 1/4$ and $n\varepsilon^2 = k^2$. Theorem 1.1 supplies $H \subseteq A + A$ with $\text{codim } H \leq 1 + 256ek^2 \leq c_k$. $\square$

For clarity, we relate this conclusion to the exact formal target. For $\alpha \leq 1$, let $g(n, \alpha)$ denote the minimum, over all $A \subseteq \mathbb{F}_2^n$ with $\text{dens}(A) \geq \alpha$, of the largest dimension of an affine subspace contained in $A + A$, taking this maximum to be zero when the sumset is empty. This is the finite-dimensional quantity called `Green51.guaranteedMaxCosetDim` in Formal Conjectures [1]. There is at least one admissible A , since $A = G$ is admissible. A linear subspace is an affine subspace, so Corollary 1.2 implies, with the same c_k, N_k ,

$$n \leq g(n, \alpha) + c_k \quad \text{whenever } n \geq N_k, \quad \frac{1}{2} - \frac{k}{\sqrt{n}} < \alpha \leq 1. \quad (24)$$

Thus the required eventual assertion follows. The strict inequality for α in the formal target is weaker than the density assumption permitted by Corollary 1.2.

7. LEAN FORMALIZATION AND REPRODUCIBILITY

The companion file `Solution.lean` contains the complete proof, including all auxiliary lemmas, in 1,925 lines. It imports the pinned Green 51 problem file and `TaskSupport`. The final declaration is `Bounty.target`; its type is obtained from `fcTypeOfName% "Green51.green_51.one_half"`. Unfolding that task type gives the proposition

$$\begin{aligned} \text{True} \iff & \left[\forall k > 0 \exists c \in \mathbb{N} \forall^\infty n \in \mathbb{N} \forall \alpha \in \mathbb{R}, \right. \\ & \left. \frac{1}{2} - \frac{k}{\sqrt{n}} < \alpha \leq 1 \implies n \leq g(n, \alpha) + c \right]. \end{aligned} \quad (25)$$

Here $\forall^\infty n$ denotes eventual truth in `Filter.atTop`. The proof establishes the right-hand side with the explicit witnesses in (2); the outer equivalence with `True` is then immediate. The task's default answer mode is used, and the unmodified task inspector confirms that the resulting target is exactly the specified one.

The formal proof follows the argument above. Its central declarations appear below. They belong to the namespace `Bounty.Green51Proof`.

Mathematical step	Lean declaration
Lemma 2.1	<code>cube_separator_variance</code>
Lemma 3.1	<code>exists_low_degree_separator_coefficient</code>
Proposition 4.1	<code>exists_good_cut</code>
Proposition 5.3	<code>small_cover</code>
Quantitative sumset bound	<code>sumset_codimension_bound</code>
Eventual assertion (24)	<code>green_one_half</code>

The quantitative sumset declaration records an affine dimension bound by constructing the common linear kernel from Theorem 1.1. The companion `PaperTheorem.lean` explicitly states the existence of this linear subspace with the claimed codimension bound.

Local compilation used Lean 4.33.1 with the following pinned revisions:

Formal Conjectures: 8432eac998110a563e03df65a28c117e97c8c142

Mathlib: 0df444a360eaa60ab8c11dca51a86af692955474

All three companion Lean files compile. The target type matches the task manifest, and the static submission checks pass. A transitive dependency audit finds no `sorryAx` and no use of the source theorem `Green51.green_51.one_half`. Lean reports only `propext`, `Classical.choice`, and `Quot.sound`. The file `CheckGreen51.lean` checks the explicit proposition and prints its axiom dependencies. Source hashes and full local verification results are recorded in the companion report.

In the matching task environment, using these dependencies and the default answer mode, place the companion files at the project root and run:

```
lake env lean -o Solution.olean Solution.lean
lake env lean CheckGreen51.lean
lake env lean PaperTheorem.lean
```

These are local checks; external submission acceptance is a separate process.

ACKNOWLEDGMENTS AND AI ASSISTANCE

The authors thank Grant Huffman (huffma97@purdue.edu) for a helpful insight.

The mathematical argument, Lean formalization, and manuscript were developed with substantial assistance from OpenAI Codex. This assistance included proposing and revising proof ideas, constructing and debugging the formal proof, running local verification, and drafting and typesetting the paper. The AI contribution thus included substantive mathematical work. Detailed local verification records accompany the formal proof.

REFERENCES

- [1] Google DeepMind, *Formal Conjectures*, source file `FormalConjectures/GreensOpenProblems/51.lean`, revision pinned in Section 7, <https://github.com/google-deepmind/formal-conjectures>.
- [2] B. Green, *100 open problems*, Problem 51, <https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf>. Accessed September 2026.
- [3] T. Sanders, *Green's sumset problem at density one half*, *Acta Arith.* **146** (2011), no. 1, 91–101. doi:10.4064/aa146-1-6. arXiv:1003.5649.

JENSEN KOHLMAYER

Email address: jenwin@purdue.edu

LIAM KRUER

Email address: kruer1@purdue.edu